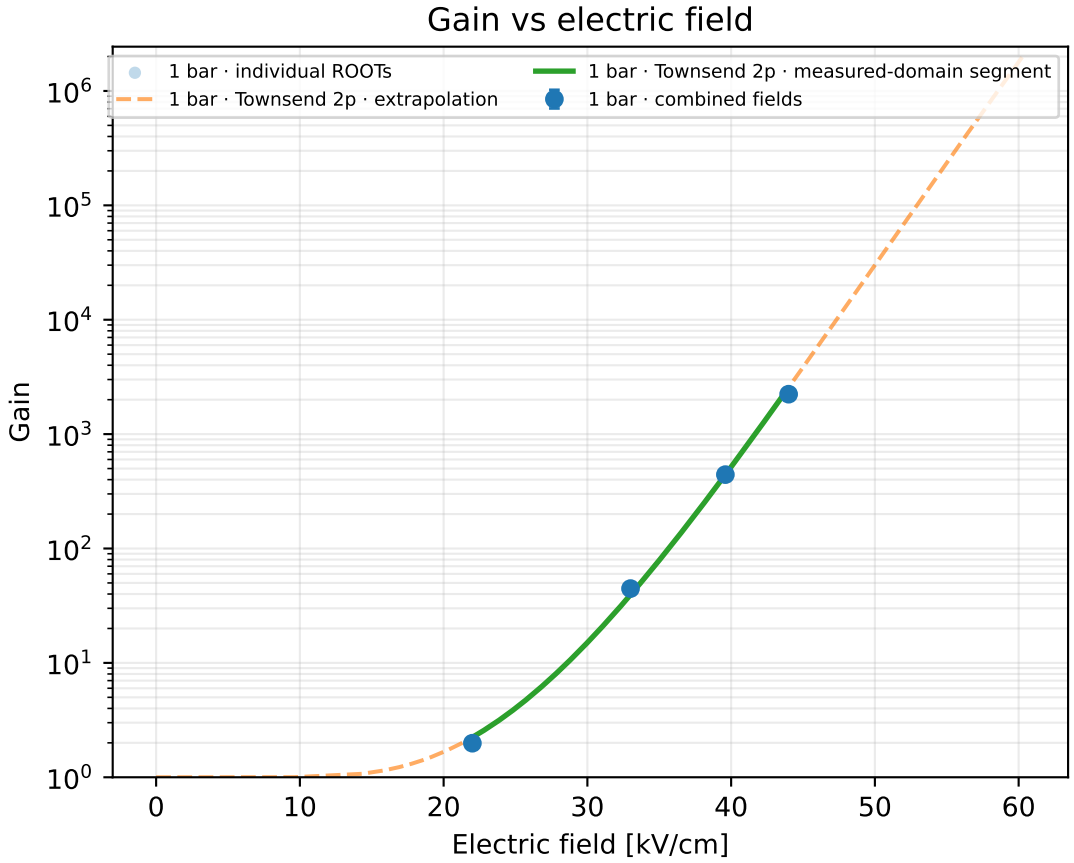
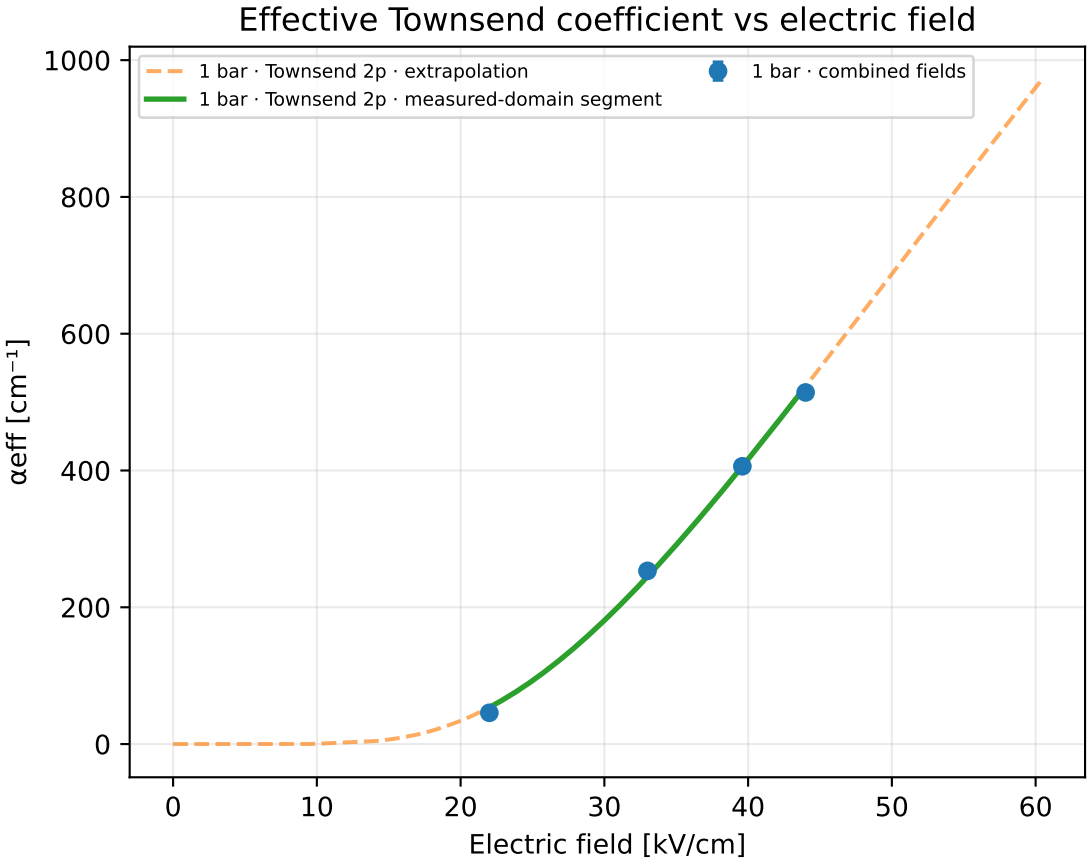
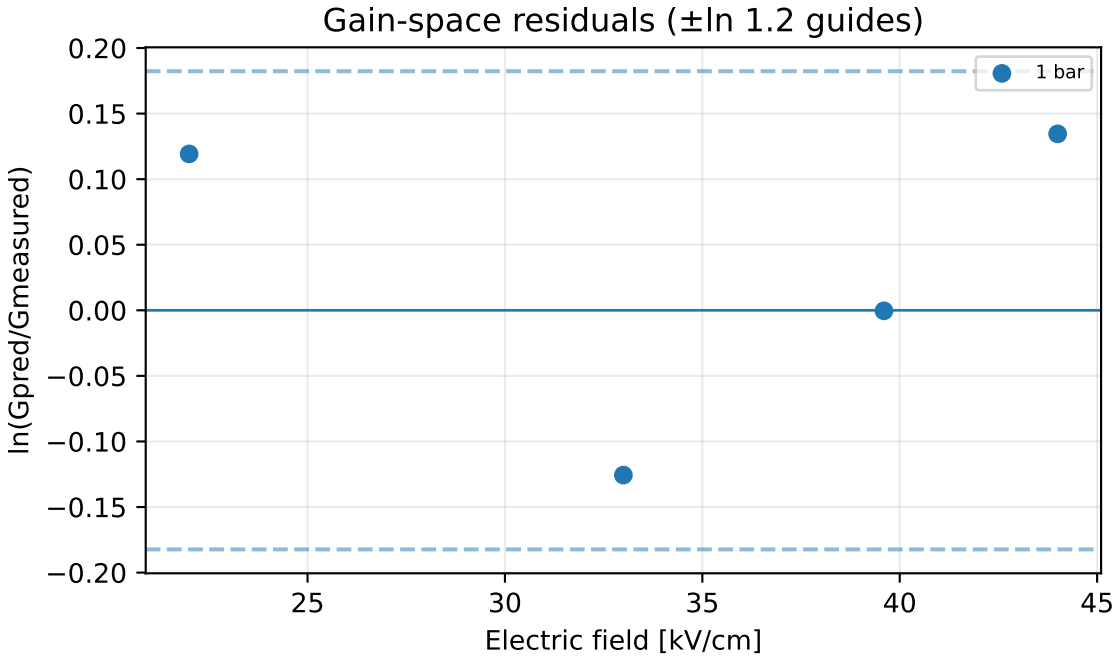
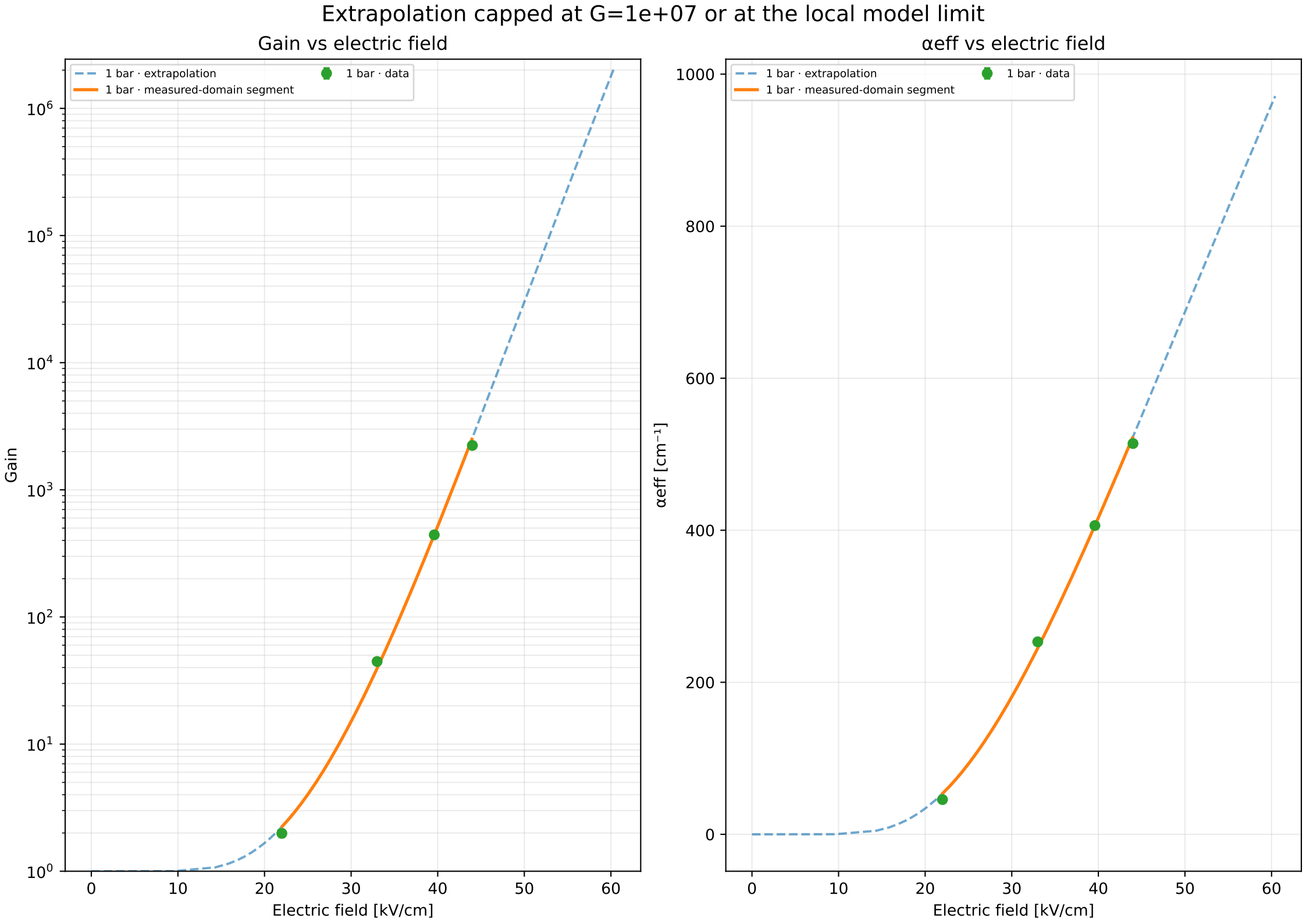
ArCF4 · Ar 80 % / CF_4 20 % · gap 0.15 mm








```
ROOT runs: 8
Unique physical points: 4
Pressures: 1 bar
Selected model: Townsend 2p
Status: rejected_poor_cross_validation
Training RMSE: ×1.086
Cross-validation median: unavailable
Cross-validation worst: unavailable
Reduced-field range: 22–44 kV cm⁻¹ bar⁻¹
A=5098.9, B=100.19, m=0, n=1
Covariance condition: 642
Warnings: poor_cross_validation, large_cross_validation_outlier
1 bar display limit: G=2.12e+06 (local inversion limit)
```


Extrapolation capped at G=1e+07 or at the local model limit
Gain vs electric field
αeff vs electric field
1 bar · extrapolation
1 bar · measured-domain segment
1 bar · data
Gain
αeff [cm⁻¹]
Electric field [kV/cm]
Electric field [kV/cm]

Exact combined points used by the fit · page 1/1

Repeated ROOTs at the same field are pooled once; all source ROOTs remain on disk.

p [bar]	E [kV/cm]	E/p	Gain	σGain	αeff [cm⁻¹]	npe	runs
1	22	22	1.985	0.138	45.7079	200	2
1	33	33	44.635	2.95	253.235	200	2
1	39.6	39.6	442.515	23.9	406.165	200	2
1	44	44	2233.715	109	514.095	200	2

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	-3.97	—	—	no	poor cross validation, large cross validation outlier